

The National Solar Radiation Data Base (NSRDB)

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ABSTRACT

The National Solar Radiation Data Base (NSRDB), consisting of solar radiation and meteorological data over the United States and regions of the surrounding countries, is a publicly open dataset that has been created and disseminated during the last 23 years. This paper briefly reviews the complete package of surface observations, models, and satellite data used for the latest version of the NSRDB as well as improvements in the measurement and modeling technologies deployed in the NSRDB over the years. **The current NSRDB provides solar irradiance at a 4-km horizontal resolution for each 30-min interval from 1998 to 2016 computed by the National Renewable Energy Laboratory's (NREL's) Physical Solar Model (PSM) and products from the National Oceanic and Atmospheric Administration's (NOAA's) Geostationary Operational Environmental Satellite (GOES), the National Ice Center's (NIC's) Interactive Multisensor Snow and Ice Mapping System (IMS), and the National Aeronautics and Space Administration's (NASA's) Moderate Resolution Imaging Spectroradiometer (MODIS) and Modern Era Retrospective analysis for Research and Applications, version 2 (MERRA-2). The NSRDB irradiance data have been validated and shown to agree with surface observations with mean percentage biases within 5% and 10% for global horizontal irradiance (GHI) and direct normal irradiance (DNI), respectively.** The data can be freely accessed via <https://nsrdb.nrel.gov> or through an application programming interface (API). During the last 23 years, the NSRDB has been widely used by an ever-growing group of researchers and industry both directly and through tools such as NREL's System Advisor Model.

1. Introduction

Understanding long-term spatial and temporal variability of the solar resource is fundamental for energy policy decisions, the optimal design of solar energy conversion systems, transmission interconnection planning, power systems integration, market operations, and reducing

uncertainty in investments [1–4]. Historical solar resource data for those purposes can be provided by ground-based in situ measurements or satellite remote sensing [3].

Pyranometers and pyrhemometers, which use either thermoelectric or photoelectric detectors, are the most common ground-based radiometers to measure global horizontal irradiance (GHI) and direct normal

Abbreviations: %RMSE, Percentage RMSE; ACS, American Cancer Society; AERONET, Aerosol Robotic Network; AOD, Aerosol Optical Depth; API, Application Programming Interface; ARM, Atmospheric Radiation Measurement; ASHRAE, American Society of Heating Refrigerating and Air-Conditioning Engineers; AVHRR, Advanced Very High Resolution Radiometer; BND, Bondville, Illinois; CPR, Clean Power Research; DHI, Diffuse Horizontal Irradiance; DISC, Direct Insolation Simulation Code; DISORT, Discrete Ordinates Radiative Transfer; DNI, Direct Normal Irradiance; DOE, Department of Energy; DRA, Desert Rock, Nevada; FARMS, Fast All-sky Radiation Model for Satellite applications; FPK, Fort Peck, Montana; GCM, Goodwin Creek, Mississippi; GEOS-4, Goddard Earth Observing System Version 4; GHI, Global Horizontal Irradiance; GIS, Geographic Information System; GOES, Geostationary Operational Environmental Satellite; GUM, Guide to the Expression of Uncertainty in Measurement; HDF5, Hierarchical Data Format; HPC, High-Performance Computing; IMS, Interactive Multisensor Snow and Ice Mapping System; ISCCP, International Satellite Cloud Climatology Project; ISD, Integrated Surface Database; MAE, Mean Absolute Error; MBE, Mean Bias Error; MERRA-2, Modern Era Retrospective analysis for Research and Applications, version 2; METSTAT, Meteorological Statistical; MODIS, Moderate Resolution Imaging Spectroradiometer; MPE, Mean Percentage Error; NASA, National Aeronautics and Space Administration; NCEI, National Center for Environmental Information; NIC, National Ice Center; NOAA, National Oceanic and Atmospheric Administration; NREL, National Renewable Energy Laboratory; NSRDB, National Solar Radiation Data Base; NWS, National Weather Service; PATMOS-x, Pathfinder Atmospheres-Extended; PSM, Physical Solar Model; PSU, Pennsylvania State University, Pennsylvania; PV, Photovoltaic; RMSE, Root Mean Square Error; RRTM, Rapid Radiative Transfer Model; SAM, System Advisor Model; SGP, Southern Great Plains; SOLRAD, Solar Radiation Network; SRB, Surface Radiation Budget; SUNY, State University of New York at Albany; SURFRAD, Surface Radiation Budget Network; SXF, Sioux Falls, South Dakota; TBL, Boulder, Colorado; TMY, Typical Meteorological Year; TOA, Top of the Atmosphere

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irradiance (DNI), respectively [5,6]. The accuracy of measurements by these instruments is highly dependent on instrument design, hardware installation schemes, data acquisition methods, and calibration method and frequency [7–9]. Measurements by accurately calibrated and well-maintained pyrheliometers and pyranometers can provide reliable long-term solar radiation data at specific locations [10–13] that are frequently used for cloud and radiation studies and validation of satellite-derived solar radiation [14–17].

The high cost of operating quality ground stations has resulted in existing surface radiation networks being sparsely distributed and insufficient to meet the needs of the rapidly growing solar energy industry. The other reliable and practical option is to use information from geostationary weather satellites that provides continuous solar radiation estimates covering a wide spectrum of temporal and spatial scales. To retrieve solar radiation from satellite data, solar irradiance models are essential to compute surface GHI and DNI from observations of radiances at the top of the atmosphere (TOA). During the last few decades, numerous solar irradiance models have been developed using empirical, semi-empirical or physical models [18–21]. Empirical models develop regression functions relating long-term GHI measurements at selected local stations to the simultaneous data recorded by satellites' visible channels [22–25] which are then used to simulate GHI from global satellite observations. The GHI is combined with empirical relationships developed using modeled or observed solar radiation to retrieve DNI [26,27]. A well-known solar radiation dataset developed by an empirical model is HelioClim based on the observations of Meteosat geostationary satellites covering Europe, Africa, the Mediterranean Basin, the Atlantic Ocean, and part of the Indian Ocean [22,28,29]. Compared to empirical models, semi-empirical models use a hybrid approach to derive solar radiation in which clear-sky background irradiance is solved from simple radiative transfer schemes [25,30,31]. A cloud index representing the proportion of radiation reflecting back to the satellite is converted to a clearness index that represents the proportion of incident radiation reaching the surface. The clearness index scales the clear sky radiation to estimate the GHI and then partitioned to estimate the DNI, which is similar to the empirical models. This semi-empirical approach has been widely implemented in global solar radiation datasets, including SolarGIS [30,31] and SolarAnywhere [25].

Physical models are conventionally categorized by single-step and two-step models according to the procedures to determine solar radiation [3]. Single-step models directly solve for GHI using satellite observations and radiative transfer theory [32–34]. Two-step models intend to understand the complete physics affecting the transmission of solar radiation from the TOA to land surface. They retrieve aerosol, cloud and other atmospheric properties from various satellite channels or modeling efforts and use the information to precisely simulate GHI by solving the radiative transfer equation [18–21]. A typical product of a two-step model is the National Aeronautics and Space Administration's (NASA's) global Surface Radiation Budget (SRB) in which International Satellite Cloud Climatology Project (ISCCP) pixel-level data and Goddard Earth Observing System Version 4 (GEOS-4) reanalysis products are used to infer atmospheric properties at a 250-km resolution every 3 h. The solar radiation is then derived using the atmospheric properties and a model developed by Pinker and Laszlo [35].

Compared to empirical, semi-empirical and single-step physical models, most two-step physical models require significant computational capability because extensive information from satellite observations and other ancillary inputs need to be processed to estimate solar radiation. The multiple processes in the production chain require sufficient quality inputs to make full use of the advanced models to reduce uncertainties of GHI and DNI. During the years, the rapid development of satellite technologies and modeling capabilities (e.g. the availability of Moderate Resolution Imaging Spectroradiometer (MODIS) spectral channels and multi-channel geostationary satellites) have effectively increased the reliability and accuracy of the two-step physical models

[3,36–39]. More recently the expansion of spectral channels with better temporal and spatial resolutions on the third-generation Geostationary Operational Environmental Satellite-16 (GOES-16; previously GOES-R) is expected to lead to remarkable improvements in aerosol and cloud products [40] which the two-step physical models are capable of exploiting. The improvements in reanalysis data, such as NASA's Modern Era Retrospective analysis for Research and Applications, version 2 (MERRA-2), bring observations and numerical models together in a unified standardized framework, resulting in high-quality ancillary information that significantly enhances the quality of the two-step physical models [41]. In contrast, empirical, semi-empirical and single-step physical models are not expected to reap equivalent benefits from advances in satellite technology and reanalysis datasets because of inherent limitations in the underlying methods.

The National Renewable Energy Laboratory (NREL) has an extensive history of developing solar resource data over the United States using various sources of observations and modeling tools. This paper reviews the evolution of NREL's National Solar Radiation Data Base (NSRDB) and the recent efforts on developing the Physical Solar Model (PSM) and satellite-based solar radiation to enhance the resolution and accuracy of the NSRDB. The remainder of this paper is structured as follows. Section 2 provides a historical review of the NREL's NSRDB. Section 3 introduces the technical details of the PSM and data validation using surface-based solar radiation measurements. Section 4 describes the users and applications of the NSRDB, and the last section concludes and explores future work to further improve the NSRDB.

2. A historical review of the NSRDB

The NSRDB is one of the most accessed public datasets providing a serially complete collection of solar energy and meteorological data, including the three most common measurements of solar radiation: GHI, DNI, and diffuse horizontal irradiance (DHI), which have been collected over the United States and a growing list of international locations with high temporal (30 min) and spatial (4 km) resolutions to accurately represent the global and regional solar radiation climates. It supports the U.S. Department of Energy's (DOE's) SunShot goals of reducing barriers to high-penetration levels of solar energy technologies by providing easy access to high-quality, foundational data that are essential for innovative product development and downstream modeling. There have been substantial improvements in data collection and modeling technologies throughout the NSRDB's more than 20 years of existence. Therefore, NREL implemented major updates to the original database three times in 2007, 2012, and 2017. The NSRDB versions are briefly reviewed below.

The first version of the NSRDB, covering 1961–1991, originated in 1994 to replace the SOLMET/ERSATZ dataset developed by the National Oceanic and Atmospheric Administration (NOAA) and DOE [47]. This version contains hourly solar irradiance data for locations over 239 ground stations across the United States with a combination of measurements (approximately 7% of the total data) and simulations using NREL's Meteorological-Statistical (METSTAT) model [42]. Cloud observations from NOAA's National Center for Environmental Information (NCEI) Integrated Surface Database (ISD) were used as inputs to the METSTAT, and measured solar irradiances were directly obtained from the National Weather Service (NWS) solar radiation network (SOLRAD).

The first version of the NSRDB was updated in 2007 to cover 1991–2005. The major updates include 10×10 km solar irradiances from hourly GOES data and the use of an empirical model developed by the State University of New York at Albany (SUNY) [43]. The satellite-based products covered the contiguous 48 states of the United States from 1998 to 2005 while the solar irradiance data in Alaska were computed by the METSTAT model. This version of the NSRDB also provides measured (less than 1%) and modeled solar irradiances as well as other meteorological data from 1454 ground stations during 1991–2005 [43].

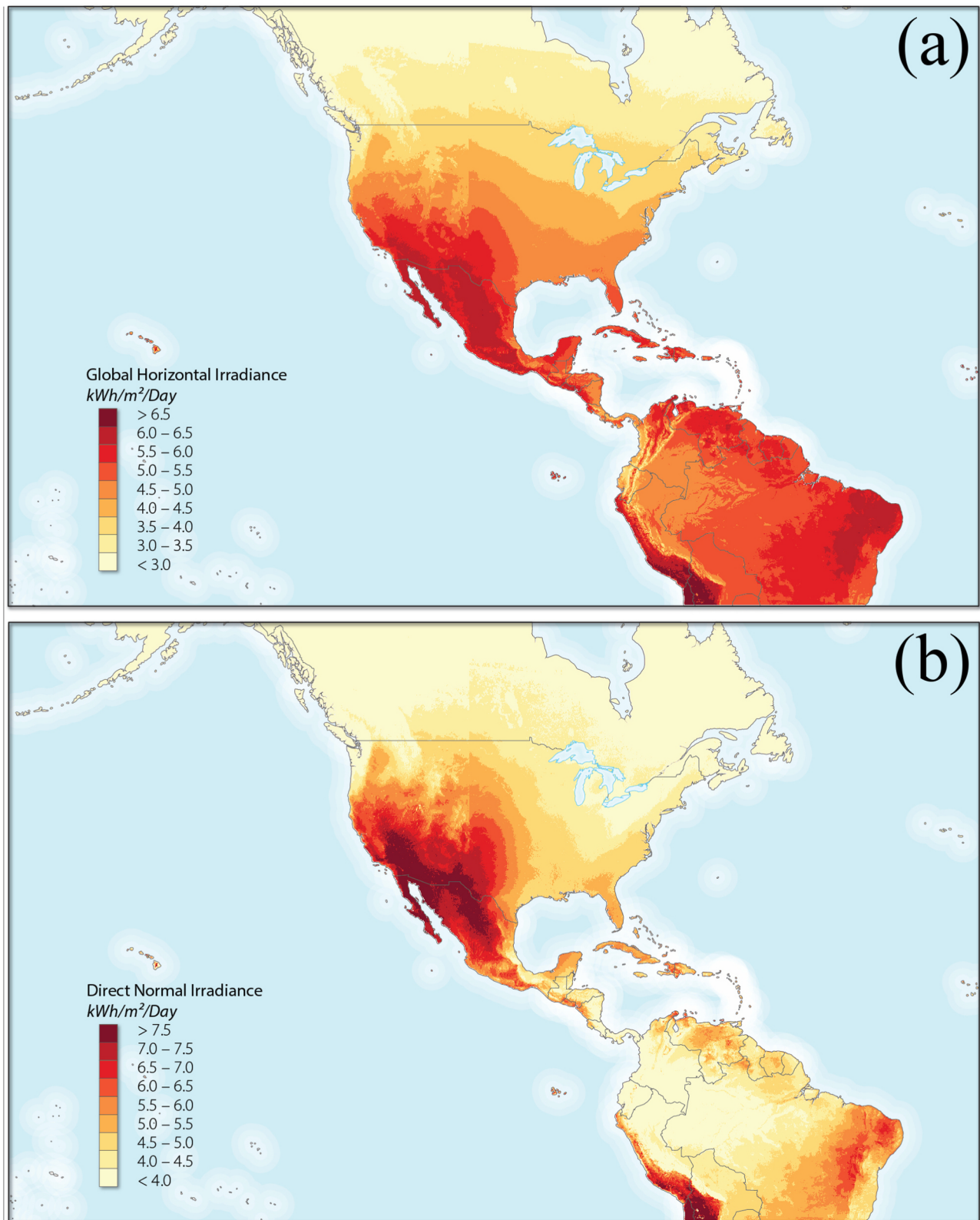


Fig. 1. Daily average of (a) GHI and (b) DNI during 1998–2016.

In 2012, NREL, in collaboration with Clean Power Research (CPR), updated the NSRDB to cover the years from 1991 to 2010. This version of the NSRDB was developed using an improved SUNY model in an hourly interval with a spatial resolution of 10×10 km. The data package also includes measurements from 1454 ground stations

(1991–1997), including meteorological data from the NCEI ISD stations. The gridded NSRDB (1998–2010) was released through NREL's Solar Prospector web portal which was decommissioned at the end of September 2016.

The latest version of the NSRDB was released in 2017 containing

gridded data from 1998 to 2016 in half-hourly temporal and 4×4 km spatial resolutions. This dataset was developed using the GOES data that cover the entire Western Hemisphere from 60° North to 20° South latitude including the contiguous United States and Central America. The average values of the daily GHIs and DNIs from 1998 to 2016 are illustrated in Fig. 1. This version used the two-step physical model, PSM, which opened the door to the use of next-generation satellite datasets for solar resource assessment and forecasting. Details about the development of this latest NSRDB are introduced below.

3. Development, validation, and delivery of the latest NSRDB (1998–2016)

With the advancement of satellite technology, information available from accurately retrieved atmospheric properties is continuously growing. This coupled with the fast advancement in computing technology has resulted in significant improvements in solar radiation simulations. One such approach is the use of two-step physical models where cloud and aerosol properties derived in the first step are fed into a radiative transfer model in the subsequent step. This approach provides the opportunity to directly calculate DNI with better accuracy from improved retrievals of water vapor, aerosol, and cloud properties [33,44,45]. NREL employed this technology to produce the latest NSRDB, which contains long-term high-resolution solar radiation from geostationary satellites.

3.1. The PSM for developing the latest NSRDB

Fig. 2 displays a flowchart of the PSM, a two-step physical model to compute solar radiation from satellite data, which was developed through collaboration among NREL, the University of Wisconsin, and NOAA. As shown in the figure, aerosol, water vapor and other meteorological properties are combined with satellite-derived cloud properties and used in the Fast All-sky Radiation Model for Solar applications (FARMS) [17] to compute GHI. For clear scenes, FARMS is also used to compute DNI, whereas the Direct Insolation Simulation Code (DISC) [27] decomposition model is used for cloudy scenes.

3.1.1. Cloud properties

NOAA developed an Advanced Very High Resolution Radiometer (AVHRR) Pathfinder Atmospheres-Extended (PATMOS-x) system to efficiently retrieve cloud physical and optical properties from the synergistic use of satellite measurements in visible, near-infrared, and infrared channels [46]. The system has been implemented with data from GOES for continuous weather monitoring and forecasting. Cloud products—including cloud height, thermodynamic phase, optical thickness and effective particle size—are retrieved from PATMOS-x and GOES-West and GOES-East satellites at 4×4 km over the continental United States for each 30 min during day time. These products are employed by the PSM to produce cloudy-sky solar radiation from 1998 to 2016.

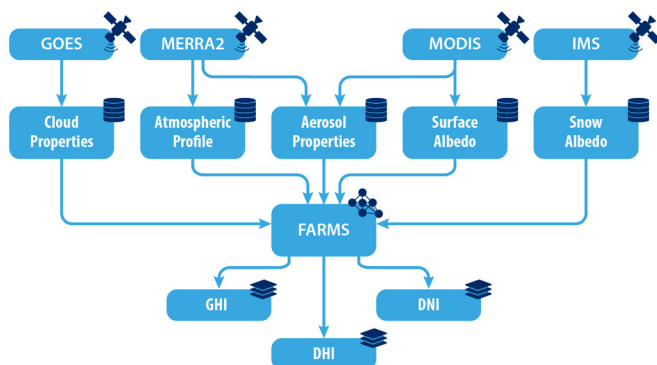


Fig. 2. A flowchart of the PSM.

3.1.2. Aerosol properties

The aerosol optical depth (AOD) used by the PSM is based on monthly MODIS data in combination with the MERRA-2 aerosol dataset. The monthly data are first scaled to a $0.5 \times 0.5^\circ$ resolution using an elevation weighting scheme and evaluated by surface-based Aerosol Robotic Network (AERONET) data. According to the evaluation, North America is divided into two regions. Over the southern and western United States, and northern Mexico, dominated by arid areas, the AOD is given by an optimal linear combination of MERRA-2 and MODIS data when the latter are available. Only MERRA-2 data are employed when MODIS observations are missing due to high surface albedo, cloudiness, large solar zenith angles, etc. Over the eastern and northern United States, Canada, southern Mexico and central America that are dominated by vegetation or urban areas, only MERRA-2 data are used because they are found to have similar accuracy as MODIS which suffers from a large fraction of missing data, especially in the high-latitude areas during winter. The monthly AOD data are then interpolated at 4×4 km on the basis of a daily average to match the NSRDB grids and improve the accuracy of surface solar radiation [47].

3.1.3. Other atmospheric and land surface data

The MODIS instruments onboard the Terra and Aqua satellites provide high-quality measurements of surface albedo at 30 arc-seconds for each 8-day interval [48]. Maclaurin et al. [49] matched the white-sky albedos from the MODIS MCD43GF product to the NSRDB grids. A point-in-polygon approach was employed to assemble MODIS pixels and compute the effective values of surface albedo within the NSRDB grids. This new product was integrated with National Ice Center's (NIC's) Interactive Multisensor Snow and Ice Mapping System (IMS) to coordinate the influence of snow and ice on surface albedo.

The other atmospheric and land properties used or provided by the NSRDB—e.g. the atmospheric profile, wind direction and speed, snow depth, surface temperature and pressure, etc.—are based on data from NASA's MERRA-2.

3.1.4. The all-sky radiative transfer model

The PSM applies satellite-derived atmospheric and land surface properties to radiative transfer models to numerically solve for solar radiation through the Earth's atmosphere. When clouds are absent, the extinction of solar radiation is associated with the light scattering by aerosols and air molecules and absorption by the trace gases in the atmosphere such as water vapor, carbon dioxide, ozone, oxygen, and methane. Although high-spectral-resolution radiative transfer models—e.g. the Line-By-Line radiative transfer model [50]—provide rigorous expression of radiation in narrow bands of wavelength, they are less efficient when solving for broadband solar radiation. Thus, many clear-sky radiative transfer models parameterize the extinction of broadband solar radiation from surface-based meteorological data or simulations in numerous spectral bands [51–57].

Badescu et al. [54] evaluated 54 clear-sky radiative transfer models using surface observations of GHI and DHI from Kipp and Zonen radiometers in Cluj-Napoca and Bucharest-Afumat. Although the best model for all scenarios was not found, REST2 [51] was ranked among the first-tier, along with three other models, in terms of the accuracy for both GHI and DHI. Because of the concise equations and parameters and the consequent efficiency in computing, REST2 is used in the development of the NSRDB [51].

The radiative transfer problem under a cloudy sky is much more complicated because of the combination of absorption and multiple scattering within the cloud. Thus, solving the radiative transfer equation for clouds is the only rigorous approach to compute cloudy-sky radiation [58,59]. Despite numerous approximations—e.g. the two-stream approach and delta-M truncation scheme [60]—conventional radiative transfer models are still time consuming in numerically solving the radiative transfer equation.

To meet the needs of developing the NSRDB and other solar energy

applications, Xie et al. [17] proposed FARMS to efficiently simulate all-sky solar radiation at land surfaces. In contrast to solving the radiative transfer equation, FARMS uses pre-computed cloud transmittances and reflectances of irradiances by the Rapid Radiative Transfer Model (RRTM) [61,62] with a 16-stream Discrete Ordinates Radiative Transfer (DISORT) model [63]. To further reduce the computing burden, the cloud transmittances and reflectances were parameterized as functions of solar zenith angle, as well as cloud thermodynamic phase, optical thickness, and particle size. The parameterization is coupled with surface albedo and REST2 accounting for clear-sky transmittances and reflectances to compute all-sky downwelling solar irradiances. The evaluation using 16-stream RRTM and DOE's Atmospheric Radiation Measurement (ARM) Southern Great Plains (SGP) site indicates that FARMS is as accurate as the two-stream approach; however, FARMS is approximately 1000 times faster than the two-stream approach, which has substantially accelerated the computation of the NSRDB. More detailed algorithm and performance evaluation can be found in [17]. It is also worth noting that the algorithm to compute clear-sky radiation is consistent in both clear-sky and cloudy-sky conditions because FARMS is coupled with REST2.

3.2. Data processing

Despite the progressive efficiency provided by the FARMS, developing the NSRDB is still a computationally intense process because of its advanced temporal and spatial resolutions and a large volume of input data including approximately 50 terabytes of GOES and 1.5 terabytes of the other data. Therefore, the NSRDB data are produced by the NREL's flagship high-performance computing (HPC) system that is capable of 2.26 PetaFLOPS with a total of 58,752 Intel Xeon processor cores, including 6912 E5-2670 SandyBridge, 24192 E5-2695v2 IvyBridge, and 27648 E5-2670v3 Haswell processor cores. To ensure timely production, allowing for rapid computation and quality check, we use Hierarchical Data Format (HDF5) for storage and a highly parallel and vectorized framework based on Python with the fundamental packages of NumPy and MPI4Py.

In the PSM shown in Fig. 2, the development of a serially complete NSRDB with consistent spatial and temporal mapping requires four additional steps prior to the computation by FARMS: regridding, temporal interpolation, time shifting, and gap filling.

The regridding step organizes all input data and reprocesses them into the NSRDB grids with a resolution of 4×4 km. The cloud properties are regridded using a nearest-neighbor approach because the GOES data have grids that are very similar to the NSRDB. Different regridding procedures based on physics laws are employed in the MERRA-2 and AOD data when their spatial resolutions are significantly lower than the NSRDB. For example, data with land-surface pressure and temperature with a 0.5° spatial resolution are regridded using an elevation scaling by considering the hydrostatic equation and a temperature lapse rate of $6^\circ\text{C}/\text{km}$, respectively. The 0.5° resolution AOD data are first reduced to sea-level using an exponential scale height of 2950 m and then regridded to the NSRDB resolution by applying the same scale height to the pixel elevation. However, the specific humidity, wind speed, and direction are converted to the NSRDB resolution using a nearest-neighbor approach.

The temporal-interpolation step assigns the regridded data to the NSRDB intervals every 30 min. Specifically, the wind directions are determined from the nearest values of the hourly MERRA-2 data, and other properties are interpolated to the NSRDB resolution using a simple linear relationship. With a mean-conserving algorithm, the monthly-mean AOD data are interpolated to daily intervals.

Data from the GOES-West are given at each integral hour and 30 min past, whereas those for the GOES-East are available at 15 and 45 min past the integral hours. To develop the NSRDB with a consistent time stamp matching the GOES-West, the time-shifting step projects the 15-min delayed cloud properties from the GOES-East data to the NSRDB

time stamps.

The gap-filling step supplements the NSRDB because data gaps in cloud properties routinely exist in long-term satellite-based observations. Times with missing cloud properties, caused by various reasons, are represented by the clear-sky GHI and the ratio of GHI to clear sky GHI available in the nearest previous time point.

3.3. Validation

A comprehensive evaluation of the NSRDB is essential when discussing “bankable data” for all phases of solar energy conversion projects, from the conceptual phase to routine solar power plant operation. Solar radiation data with known uncertainties help reduce the expense associated with mitigating performance and financing risk for solar energy conversion systems.

The performance of the latest NSRDB was recently investigated by Habte et al. [64] using surface observations from NREL's SRRL; ARM SGP; and Surface Radiation Budget Network (SURFRAD) sites at Bondville, Illinois (BND); Desert Rock, Nevada (DRA); Fort Peck, Montana (FPK); Goodwin Creek, Mississippi (GCM); Pennsylvania State University, Pennsylvania (PSU); Sioux Falls, South Dakota (SXF); and Boulder, Colorado (TBL). Various statistics, such as mean bias error (MBE), mean absolute error (MAE), mean percentage error (MPE), root mean square error (RMSE), and percentage RMSE (%RMSE) were calculated for various locations and time scales. Fig. 3 demonstrates the spatial distribution of the surface sites that represent diverse geographical and climatic features at locations throughout the continental United States.

Fig. 4 shows the MPEs of the GHI and DNI that are defined as:

$$MPE = \frac{\sum_{i=1}^n (x_N - x_S)}{\sum_{i=1}^n x_S} \times 100\% \quad (1)$$

where x_N denotes hourly-, daily-, monthly- or annually-averaged NSRDB data during 1998–2016; x_S represents the corresponding surface observations in the same timescales; and n is the number of data points for each averaging timescale. The biases of the NSRDB do not significantly vary with the averaging timescales, which indicates that its performance is consistent when used for different phases of solar energy deployment projects. The GHIs from the NSRDB are generally over- and underestimated, respectively, in the eastern and western United States; whereas the overall percentage biases are within 5%. The MPEs of the DNIs are within 10% on average, whereas those over the eastern United States are significantly higher than the western area, because of the presence of more clouds. However, challenges exist in accurately quantifying irradiances in the western areas that are susceptible to high occurrences of clouds, snow, and bright surfaces, such as FPK and TBL. The uncertainty and bias of DNI is generally greater than GHI because DNI is directly related to extinction in the atmosphere including clouds and aerosols whereas GHI includes radiation that is scattered to the surface [57]. Thus, advanced technology in satellite remote sensing of clouds and improvements in radiative transfer models are needed to improve the accuracy in the DNI simulation. Sub-pixel variability and the parallax effect in clouds contribute to increased uncertainty and biases for both GHI and DNI, which will be investigated in future updates of the NSRDB.

It was also reported by Habte et al. [64] that the %RMSEs of the hourly-averaged GHI and DNI can reach up to 20% and 40%, respectively, when compared to the surface-based measurements. An inter-annual variability of the GHI and DNI was found to be less than 5% for both the NSRDB and surface observations.

It is known that uncertainties in ground-based measurements might affect the evaluation of biases from satellite data. For a more comprehensive analysis, we evaluated the NSRDB using a quantification method based on the Guide to the Expression of Uncertainty in Measurement (GUM) [65], which implements the effect of statistical

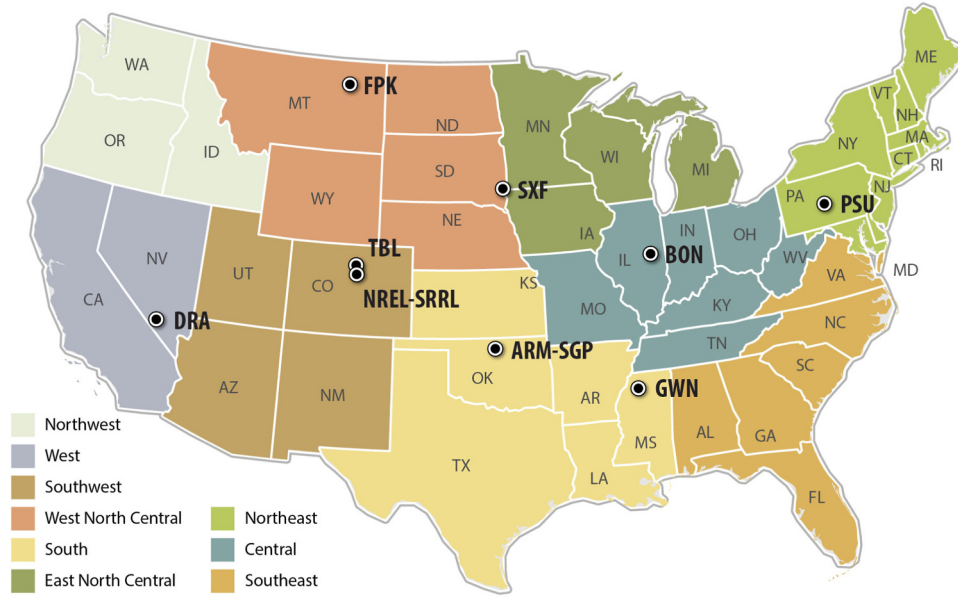


Fig. 3. The spatial distribution of NREL SRRL, ARM SGP, and SURFRAD sites.

distributions when comparing the model simulations and surface observations, accounts for the uncertainties in surface observations, and represents the model biases with a confidence interval. Eq. (2) denotes the percentage bias for a 95% confidence interval:

$$U_{95} = \pm \sqrt{U_s^2 + MPE^2 + \%RMSE^2} \quad (2)$$

where U_s is the percentage bias of the surface measurements and $\%RMSE$ is defined by

$$\%RMSE = \sqrt{\frac{\sum_{i=1}^n (x_N - x_S)^2}{\sum_{i=1}^n x_S^2}} \times 100\% \quad (3)$$

The magnitudes of U_{95} for the GHI at the surface sites are illustrated in Fig. 5. A U_s of 5% is assumed for all the sites because most well-maintained surface-based pyranometers report uncertainties ranging from 3% to 5% [8,66]. The stability of U_s and its impact on the U_{95} might require further studies, as discussed by [64]. Fig. 5 shows that the

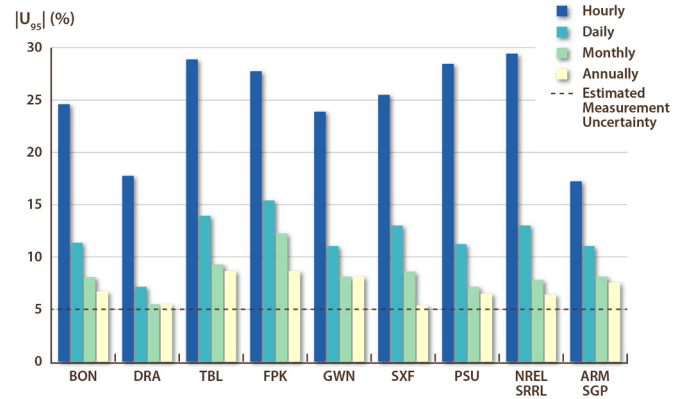


Fig. 5. The magnitudes of U_{95} for the hourly-, daily-, monthly-, and annually-averaged GHIs. The percentage bias of the surface measurements is assumed to be 5% for all the sites.

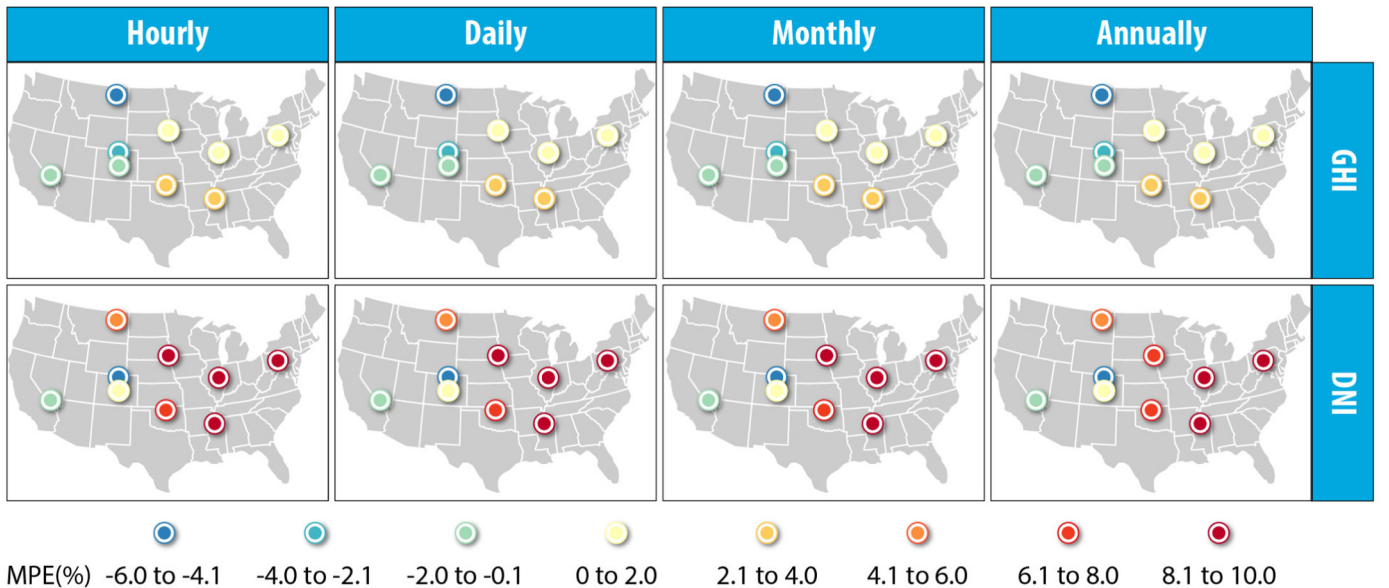


Fig. 4. The MPEs of the hourly-, daily-, monthly-, and annually-averaged GHI and DNI during 1998–2016.

magnitude of U_{95} dramatically decreases when the averaging timescale varies from hourly to annually. Therefore, %RMSE plays a dominant role in understanding the uncertainty of hourly-averaged GHI whereas the bias from surface observations becomes important in the monthly- and annually-averaged solar radiation. More details in the validation of the NSRDB can be found in [64].

3.4. Dissemination

Geographic Information System (GIS) layers are the core mechanism for developers, transmission planners, or conservationists to display geographic datasets in multiple geospatial processing programs. The GIS layers present in the NSRDB data include annual and multi-year mean GHI and DNI and multi-year mean capacity factors modeled for single-axis tracking photovoltaic (PV) panels and those with a fixed tilt angle of 20°. All the GIS layers are downloadable in several formats including the commonly used shapefile.

The NSRDB Viewer (<https://nsrdb.nrel.gov/nsrdb-viewer>), demonstrated in Fig. 6, is the major data delivery tool. Built on the NREL's OpenCarto, a web-based GIS framework, the NSRDB Viewer provides an intuitive, map-based interface for accessing raw time-series data or summarized GIS layers. Detailed instructions for downloading data from the NSRDB Viewer can be found on the website.

The NSRDB also provides an Application Programming Interface (API) (<http://developer.nrel.gov/docs/solar/nsrdb>) giving researchers, analysts, and developers an alternative way to efficiently download data using modern scripting languages, e.g., Python, MATLAB, or R. In addition, the API enables web developers to build their own applications without storing the approximately 50 terabytes of the NSRDB data.

4. NSRDB users and applications

Because of the improved accuracy and availability, the latest NSRDB has become a heavily and increasingly used dataset since its deployment. According to the web-based counter, the monthly data visit from the NSRDB viewer has doubled to more than 10,000 in 12 months, whereas more than 40% of the visit was from unique users. The NSRDB users include universities, local and federal governments, research institutes, public utilities, and numerous energy and high-technology companies across the world. Major use of the NSRDB can be categorized by energy-related and other applications.

The energy-related applications of the NSRDB include, but are not limited to, site and building design, facility integration, transmission and distribution planning, and strategic analysis. The serially completed, spatially continuous solar radiation from the NSRDB naturally meets the demands in developing solar radiation time-series [67] and Typical Meteorological Year (TMY) data for building analysis and the forecast and comparison of solar system performance [68–71]. In addition, capacity expansion and integrated assessment models rely on the spatially continuous NSRDB data to quantify the supply and quality of solar power and assess costs and feasibilities at a national scale. Production cost models used in grid integration studies to evaluate and optimize power plant dispatch require the high temporal-resolution NSRDB data from thousands of locations. Models that are intrinsically coupled with capacity expansion and production cost models—known as Geodesign models—are used to characterize and quantify solar supply. Those models use continuous or climatological data based on the NSRDB to evaluate land-use impacts, barriers, and scenarios of development futures. Solar energy facility developers use PVSyst or System Advisor Model (SAM) with long-term NSRDB data to estimate power output and assess specific cost and feasibility. A brief summary

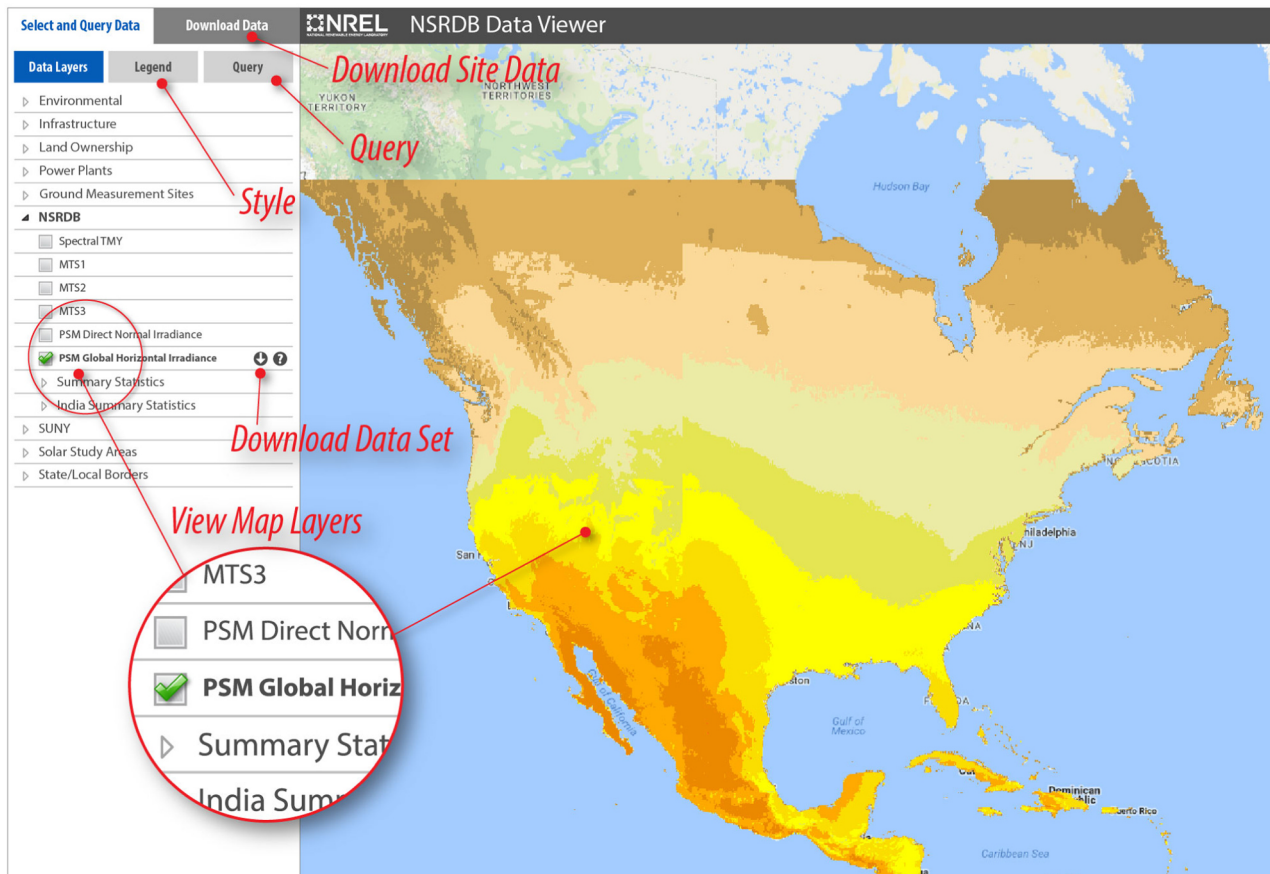


Fig. 6. A screenshot of the NSRDB Viewer.

Table 1
Models developed using the NSRDB.

Model	Description	NSRDB data	References
EnergyPlus	A whole building energy simulation program to model energy consumption and water use in buildings	TMY or historical solar radiation	[73]
HAIKU	A simulation model of regional electricity markets and interregional electricity trade in the continental United States	TMY or historical solar radiation	[74]
IPM	A multi-regional, dynamic, deterministic linear programming model of the U.S. electric power sector	Historical solar radiation	[75]
MARKAL	A generic model to represent the evolution during a period of 40–50 years of a specific energy system at a national, regional, state/province, or community level	TMY or historical solar radiation	[76]
NEMS	An integrated model of the U.S. energy system linked to a macroeconomic model	TMY or historical solar radiation	[77]
PLEXOS	A commercial production cost model	Historical solar radiation	[78,79]
PVSyst	A performance model for architecture, engineering, and education	TMY or historical solar radiation	[80]
PVWatts	For estimating the energy production of a grid-connected PV system	TMY or historical solar radiation	[81]
ReEDS	An optimization model of U.S. electricity sector with special focus on issues relating to renewable energy technologies	TMY or historical solar radiation	[82]
RPM	A capacity expansion model with high spatial and temporal resolution that can be used for mid- and long-term scenario planning of regional power systems	Historical solar radiation	[83]
SAM	A performance and financial model designed to facilitate decision making for people involved in the renewable energy industry	TMY or historical solar radiation	[84]
SIND Toolkit	For regional solar generation integration studies by providing modeled, coherent sub-hourly solar power data, information, and tools	1-min solar radiation from a statistical up sampling of the 30-min NSRDB.	[85]
SolarDS	A distributed solar capacity expansion model to evaluate scenarios in the U.S.	Historical solar radiation	[86]

and description of the abovementioned models is provided in Table 1. The NSRDB has also been used in bioenergy to evaluate algal biomass productivity potential in a variety of climatic zones [72].

In addition to the energy-related applications, the NSRDB has been employed in many other research areas. For example, the American Society of Heating Refrigerating and Air-Conditioning Engineers (ASHRAE) uses the NSRDB for climate research. The NSRDB has also been used by the American Cancer Society (ACS) to conduct cancer research because solar exposure is the primary vitamin D source that is associated with survival in multiple cancers. The residence-based ultraviolet radiation data from the NSRDB is used to examine its relationship to cancer outcomes and help understand the geographic disparities in cancer prognosis.

5. Conclusions and future plans

The NSRDB is a widely used public solar resource dataset that has been developed and updated during more than 20 years to reflect advances in solar radiation measurement and modeling. The most recent version of the NSRDB uses 30-min satellite products at a 4×4 km resolution that cover the period 1998–2016. The NREL-developed PSM was the underlying model for developing this recent update, which used this two-step physical model and took advantage of the progressive computing capabilities and high-quality meteorological datasets from NOAA's GOES; NIC's IMS; and NASA's MODIS and MERRA-2 products. The percentage biases in the latest NSRDB are approximately 5% for GHI and approximately 10% for DNI when compared to the long-term solar radiation observed by the ARM, NREL, and SURFARD stations across the United States.

Future updates of the NSRDB are expected annually. Advanced information in the planned dataset will involve new satellite retrievals and improved AOD data. However, future advancements in the PSM—e.g. identifying low clouds and fog in coastal areas, improving the discrimination of clouds from snow, providing specular reflection on bright surface, and reducing uncertainties of parallax especially under high-resolution conditions—are desired to further increase the accuracy of the NSRDB. Further, the Lambert-Bouguer Law [59] is almost non-exclusively utilized by physics-based radiative transfer models, including FARMS, that assume DNI is constituted of an infinite narrow beam. This assumption is interpreted differently in surface-based observations by pyrheliometers where direct solar radiation is defined as the “radiation received from a small solid angle centered on

the sun's disc” [87]. To reduce this disagreement in principle, we employed an empirical model, DISC [27], to decompose DNI from the GHI in cloudy situations. Further efforts are underway in developing a new DNI model to bridge the gap between model simulation and surface observation. Additionally, the launch of GOES-16 is also expected to provide improved cloud products; however, this requires better capabilities to process larger volumes of data. Finally, while the PSM has been applied to the GOES satellites the methods and models are equally applicable to any other geostationary satellites. Therefore, future work will involve developing global capabilities in collaboration with various national and international partners.

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